

Closing the Amortization Gap in Neural Topic Models

via Langevin Posterior Refinement

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Abstract

We study the *amortization gap* in variational topic models—the systematic approximation error that arises when a single encoder network must approximate the posterior for every document simultaneously. We propose **EBM-LTM**, a framework that addresses this gap through *Langevin posterior refinement*: starting from the amortized encoder’s output, short-run Stochastic Gradient Langevin Dynamics (SGLD) iteratively corrects document-specific inference errors. An optional energy-based prior captures multi-modal corpus structure beyond fixed Gaussian priors.

Our central empirical finding is that *ETM+Langevin is the primary protagonist*: applying Langevin refinement to a standard ETM encoder raises NPMI from 0.0115 to 0.0248 on 20 Newsgroups (+**116%**), with the energy-based prior contributing only marginal additional gain. We evaluate on three benchmarks (20NG, IMDb, NYT) and conduct ablation studies on Langevin step count, 2×2 factorial decomposition, Langevin step size, and VICReg-style variance regularization. We also include BERTopic and FASTopic in our extended comparison, and report held-out perplexity distinguishing PPL-amortized from PPL-refined. We prove that the W_2 distance between the refined distribution and the true posterior contracts geometrically with SGLD steps under a log-Sobolev condition (Theorem 4.1), and validate this prediction across datasets via the empirical amortization gap ($\Delta \log \text{Lik}$): 20NG benefits $10\times$ more than IMDb. Posterior collapse (Active Units = 0) persists despite the AU regularizer, VICReg-style variance regularization on the latent space improves NPMI by +15% at seed=42 but shows mixed results in multi-seed evaluation: it reduces training variance by 53% (Table 7) without consistently improving mean NPMI, and AU=0 persists (Section 3.6). These results establish **Langevin refinement as a broadly applicable drop-in module** for VAE-based topic models, with measurable benefit wherever the amortization gap is large.

1 Introduction

Neural topic models based on variational autoencoders [Kingma and Welling, 2014] have achieved strong empirical results by replacing slow MCMC inference with amortized variational inference [Srivastava and Sutton, 2017, Dieng et al., 2020, Card et al., 2018, Wu et al., 2023]. However, this convenience comes at a cost: the *amortization gap* [Cremer et al., 2018].

The amortization gap. An amortized encoder $q_\lambda(\theta|x)$ must share parameters across all documents. For document x , the true ELBO gap decomposes as:

$$\log p(x) - \mathcal{L}_{\text{ELBO}} = \underbrace{\text{KL}(q^*(\theta|x) \parallel p(\theta|x))}_{\text{approximation gap}} + \underbrace{\text{KL}(q_\lambda(\theta|x) \parallel q^*(\theta|x))}_{\text{amortization gap}}, \quad (1)$$

where q^* is the optimal variational family member for this document. Cremer et al. [2018] showed that the amortization gap accounts for over half the total gap in practice.

Our approach. We propose to close the amortization gap with *per-document Langevin refinement*: the encoder provides a fast warm-start initialization, and L_v steps of SGLD correct the document-specific error afterward. This is optionally combined with an energy-based prior $p_\alpha(z) \propto \exp(-f_\alpha(z))\mathcal{N}(z; 0, I)$ that can

adapt its shape to the corpus geometry. To address the resulting posterior collapse ($AU=0$), we also propose *VICReg-style variance regularization* in latent z -space (Section 3.6), which targets the root cause of mode convergence more effectively than θ -space penalties.

Main finding. Our factorial ablation (Table 5) reveals that **Langevin refinement, not the EBM prior, drives the improvement**: ETM+Langevin achieves $NPMI = 0.0248$, nearly matching full EBM-LTM (0.0242) at lower complexity. The EBM prior adds modest topic diversity (TD +0.012). We report this result honestly and argue that it is itself valuable: *Langevin refinement is a principled, broadly applicable module that any VAE topic model can adopt.*

Contributions.

- C1. Langevin posterior refinement for topic models:** We demonstrate that per-document SGLD steps substantially reduce the amortization gap, yielding a +116% $NPMI$ improvement over ETM on 20NG (+32% on NYT), robust across 3 random seeds.
- C2. EBM prior integration:** We show how an energy-based prior can be jointly trained with contrastive divergence, adding topic diversity at the cost of additional computation.
- C3. Theory with cross-dataset validation:** We provide a convergence bound (Theorem 4.1) under a log-Sobolev inequality (LSI) condition and validate its prediction empirically: larger amortization gap ($\Delta\log\text{Lik}$) predicts larger $NPMI$ gain.
- C4. VICReg-style diversity regularization:** We propose variance-covariance regularization in z -space (Section 3.6). At seed=42, the best config achieves +15% $NPMI$, but multi-seed evaluation reveals mixed results: training variance is reduced by 53% while mean $NPMI$ is not consistently improved (Table 7). $AU=0$ persists, confirming the softmax bottleneck as the root cause.
- C5. Comprehensive evaluation:** Step-size ablation, held-out perplexity (PPL-amortized vs. PPL-refined), and a three-argument critique of BERTopic’s $NPMI$ structural bias provide a complete empirical picture of Langevin-based topic modeling.

Scope and limitations of this paper. We evaluate on three standard benchmarks (20NG, IMDB, NYT) and include BERTopic and FASTopic in extended comparison. Larger-scale corpora (Wiki, WOS) and comparisons with LLM-based topic models (TopClus) remain important future directions. We target this paper at venues where methodological novelty and honest evaluation are valued (EMNLP Findings, AISTATS, NeurIPS workshops).

2 Related Work

VAE-Based Neural Topic Models. NVDM [Miao et al., 2016] first applied VAEs to documents. ProLDA [Srivastava and Sutton, 2017] introduced a product-of-experts decoder. ETM [Dieng et al., 2020] uses embedding-space topic-word similarity. SCHOLAR [Card et al., 2018] conditions on metadata. CombinedTM [Bianchi et al., 2021] uses pretrained SBERT embeddings. ECRTM [Wu et al., 2023] enforces topic diversity via Earth-mover regularization. All share fixed priors and pure amortized inference.

Recent non-VAE topic models. More recent work has moved beyond VAEs: BERTopic [Grootendorst, 2022] uses contextual embeddings + HDBSCAN clustering; FASTopic [Wu et al., 2024a] achieves fast training via optimal transport; TopClus [Meng et al., 2022] exploits cluster structure in embedding space. We include BERTopic and FASTopic in our extended evaluation (Section 5.3), showing that their apparent $NPMI$ superiority on news corpora is partly an artifact of hard-assignment bias and collapses on short-text corpora (IMDB). TopClus requires LLM-based pseudolabels and remains a future comparison target.

Self-supervised representation regularization. VICReg [Bardes et al., 2022] prevents representation collapse in self-supervised learning via variance, invariance, and covariance terms. We adapt the variance and covariance terms to the topic model z -space (Section 3.6), operating pre-softmax to avoid the θ -space bottleneck that makes the AU regularizer ineffective.

Addressing the amortization gap. Cremer et al. [2018] quantify the gap empirically. Semi-Amortized VAEs [Kim et al., 2018] refine encoder outputs with gradient steps. Marino et al. [2018] learn iterative inference networks. Hoffman [2017] and Ruiz and Titsias [2019] study posterior MCMC refinement in latent VAEs. We are the first to systematically apply posterior Langevin refinement to neural topic models and to quantify its contribution via factorial ablation.

Energy-Based Models. LeCun et al. [2006], Du and Mordatch [2019], Deng et al. [2020] apply EBM to various domains. Nijkamp et al. [2019] establish short-run MCMC as a valid EBM training algorithm. Pang et al. [2020] pioneer latent-space EBM priors for image VAEs. Liu et al. [2025] apply an energy prior to topic models but omit posterior refinement and do not quantify the prior vs. inference contributions separately.

3 Method

3.1 Generative Model

Let $x \in \mathbb{Z}_{\geq 0}^V$ be a bag-of-words document. EBM-LTM defines:

$$z \sim p_\alpha(z) = \frac{1}{Z_\alpha} \exp(-f_\alpha(z)) \cdot \mathcal{N}(z; 0, I_K), \quad (2)$$

$$\theta = \text{softmax}(z), \quad x \sim \text{Multinomial}(N, \beta^\top \theta), \quad (3)$$

where $\beta = \text{softmax}(\rho^\top \alpha_k)$ is the ETM-style topic-word matrix [Dieng et al., 2020]. When $f_\alpha \equiv 0$, the model reduces to standard ETM with Gaussian prior.

3.2 Energy Network

f_α is a two-layer MLP with SiLU activation and spectral normalization [Miyato et al., 2018]:

$$f_\alpha(z) = \mathbf{w}^\top \text{SiLU}(W_2 \text{SiLU}(W_1 z + b_1) + b_2),$$

with hidden dimension $2K$. Spectral normalization bounds the Lipschitz constant, stabilizing both the Langevin dynamics and contrastive divergence training.

3.3 Langevin Posterior Refinement (the Key Module)

Given amortized initialization $z_0 \sim q_\lambda(\cdot|x) = \mathcal{N}(\mu(x), \sigma^2(x))$, we run L_v steps of overdamped SGLD targeting $p(z|x)$:

$$z_{t+1} = z_t - \frac{\delta_v}{2} \nabla_z [f_\alpha(z_t) - \log p_\beta(x|z_t)] + \sqrt{\delta_v} \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, I). \quad (4)$$

The encoder provides a warm start; energy network parameters are frozen during these steps to avoid unnecessary graph construction. When $f_\alpha \equiv 0$, this reduces to Langevin refinement of a Gaussian posterior using only the likelihood gradient.

Encoder imitation loss. The encoder is trained to track the Langevin output:

$$\mathcal{L}_{\text{init}} = \|z_0 - \text{sg}(z_{L_v})\|_2^2, \quad (5)$$

creating a feedback loop: as q_λ improves, fewer steps suffice.

Algorithm 1: EBM-LTM Training

Input: Dataset $\mathcal{D}$; f_α , p_β , q_λ ; $L_s=30$, $L_v=15$, $\delta_s=0.1$, $\delta_v=0.02$, warmup fraction $T_w=0.25$.

for epoch = 1, ..., T **do**

for batch $\{x_i\}_{i=1}^B \sim \mathcal{D}$ **do**

if epoch $\leq T_w T$ **then** *Warmup: standard VAE loss*

$z_0 \sim q_\lambda(\cdot|x)$; Update λ, β via $-\log p_\beta(x|z_0) + \text{KL}(q_\lambda\|p_0)$

else *Main phase*

 (1) $z_0 \sim q_\lambda(\cdot|x)$ *amortized init*

 (2) $z_{L_v} \leftarrow \text{SGLD}(z_0, x, L_v, \delta_v)$ *posterior refinement (Eq. 4)*

 (3) Update β via $-\log p_\beta(x|z_{L_v})$ *decoder step*

 (4) $z' \leftarrow \text{SGLD}(\mathcal{N}(0, I), L_s, \delta_s)$; Update α via Eq. (6) *energy CD*

 (5) Update λ via Eq. (5) *encoder imitation*

end if

end for

end for

Figure 1: EBM-LTM training. Setting $f_\alpha \equiv 0$ and $L_s = 0$ gives the computationally cheaper **ETM+Langevin** variant.

3.4 Energy Prior Training

f_α is trained by contrastive divergence:

$$\mathcal{L}_\alpha = \mathbb{E}_{\text{data}}[f_\alpha(z_{L_v})] - \mathbb{E}_{p_\alpha}[f_\alpha(z')], \quad (6)$$

where prior samples z' are obtained by L_s steps of SGLD from $\mathcal{N}(0, I)$ [Nijkamp et al., 2019]. Separate Adam optimizers are used for the energy network ($\eta_\alpha = 10^{-4}$) and remaining parameters ($\eta = 2 \times 10^{-3}$).

3.5 Training Schedule

The warmup phase (first $T_w T$ epochs as standard VAE) prevents early posterior collapse caused by the randomly initialized energy landscape. Without warmup, the energy prior produces chaotic gradients before it has learned any meaningful structure, causing the encoder to diverge (confirmed by our $L_v=0$ ablation, Table 4).

3.6 VICReg-Style Collapse Prevention

The AU=0 posterior collapse documented in Section 6.1 motivates a principled variance regularizer applied directly in z -space. We adapt VICReg [Bardes et al., 2022] to the topic model latent space.

Variance term. For a batch of latent codes $z \in \mathbb{R}^{B \times K}$, we penalize dimensions with below-target standard deviation:

$$\mathcal{L}_{\text{var}} = \frac{1}{K} \sum_{k=1}^K \max(0, \gamma - \sqrt{\text{Var}_b(z_k) + \varepsilon}), \quad (7)$$

where $\gamma = 1.0$ is the target std and $\varepsilon = 10^{-4}$ for numerical stability. This encourages every latent dimension to carry meaningful signal across the batch.

Covariance term. To prevent dimensions from encoding the same information, we penalize off-diagonal correlations in the batch covariance:

$$\mathcal{L}_{\text{cov}} = \frac{1}{K^2} \sum_{i \neq j} [\text{Cov}(z_i, z_j)]^2. \quad (8)$$

Combined regularizer.

$$\mathcal{L}_{\text{VICReg}} = \lambda_v \mathcal{L}_{\text{var}} + \lambda_c \mathcal{L}_{\text{cov}}, \quad (9)$$

added to the main training loss (both warmup and main phases). During the main phase, VICReg is applied to the Langevin-refined z_{L_v} , directly regularizing the distribution that informs the decoder. Section 5.7 presents a full grid search over λ_v and λ_c .

4 Theoretical Analysis

4.1 Langevin Refinement Under Log-Sobolev Conditions

We first state our convergence result under a *log-Sobolev inequality* (LSI) condition, which is weaker than strong convexity and more appropriate for topic model posteriors.

Theorem 4.1 (Amortization Gap Contraction). *Let $\pi(z) = p(z|x)$ satisfy a log-Sobolev inequality with constant $\rho > 0$. Let q_0 satisfy $W_2(q_0, \pi) \leq \varepsilon$. After T steps of the Langevin Markov chain (Eq. 4) with step size $\delta \leq \rho/M^2$ (where M is the Lipschitz constant of $\nabla \log \pi$), the distribution q_T satisfies:*

$$W_2(q_T, \pi) \leq e^{-\rho T \delta / 2} \varepsilon + \frac{CM}{\rho} \sqrt{\delta K}, \quad (10)$$

where K is the latent dimension and $C > 0$ is a universal constant.

Proof sketch. The LSI with constant ρ implies $\text{KL}(q||\pi) \leq \frac{1}{2\rho} \mathcal{I}(q||\pi)$. Under the Langevin dynamics, $\frac{d}{dt} \text{KL}(q_t||\pi) \leq -\rho \text{KL}(q_t||\pi)$, giving exponential decay in KL. Converting KL to W_2 via Talagrand’s T_2 inequality [Otto and Villani, 2000] gives the continuous-time bound. The Euler–Maruyama discretization introduces a bias bounded by $CM\sqrt{\delta K}/\rho$ per step [Dalalyan, 2017]. $\square$

Remark 4.2 (Gap between theory and practice for topic models). *The LSI condition is satisfied by log-concave distributions; for strongly convex potentials, $\rho \geq m$ (the strong convexity constant). However, the topic model posterior $\log p(z|x) = -f_\alpha(z) - \frac{1}{2}\|z\|^2 + \sum_w x_w \log[\text{softmax}(z)^\top \beta_w]$ may not be log-concave globally, since the log-softmax is concave but not strongly so. In practice, the warm start from q_λ places z_0 in a region where the posterior potential is approximately convex (near the mode), making the bound descriptive rather than vacuous. Our ablation (Table 4) provides direct empirical validation: NPMI improves monotonically with L_v , consistent with the bound’s prediction.*

4.2 Universal Approximation of EBM Priors

Theorem 4.3 (Universal Approximation). *For any continuous density p^* on $\mathbb{R}^K$ with finite second moment and $\epsilon > 0$, there exists a two-layer MLP energy function f_α such that $\text{KL}(p^*||p_\alpha) < \epsilon$, where $p_\alpha \propto \exp(-f_\alpha) \mathcal{N}(0, I)$.*

Proof sketch. Set $f_\alpha(z) = -\log[p^*(z)/\mathcal{N}(z; 0, I)] + C$, which gives $p_\alpha = p^*$. Since this target function is continuous, Hornik’s theorem [Hornik, 1991] provides the MLP approximation; continuity of the Gibbs map yields KL convergence. $\square$

Remark 4.4. *This result is essentially a corollary of universal approximation and is included for completeness. The practical question—whether a finite-width network trained via short-run MCMC converges to a good approximation—is not addressed by this theorem and remains an empirical matter.*

4.3 On the Marginal Role of the EBM Prior

Theorem 4.1 applies when $f_\alpha \equiv 0$ (Gaussian prior + Langevin) as a special case. Setting $f_\alpha \equiv 0$ does not change the contraction factor; it only modifies the stationary distribution being approximated. This is consistent with our empirical finding (Table 5) that ETM+Langevin achieves nearly identical NPMI to full EBM-LTM: the Langevin correction term dominates regardless of the prior. The EBM prior’s contribution is to shape the prior distribution toward the true data marginal, which manifests primarily as improved topic diversity rather than coherence.

4.4 Cross-Dataset Validation of the Convergence Bound

Theorem 4.1 predicts that the benefit of Langevin refinement scales with the initial distance $W_2(q_0, \pi)$ — the amortization gap. We now connect this prediction to our empirical measurements (Table 11).

The $\Delta\log\text{Lik}$ values in Table 11 serve as a *proxy for the amortization gap*: they measure how much the posterior log-likelihood improves when z_0 is replaced by Langevin-refined z_{L_v} . Under Theorem 4.1, this improvement should be proportional to $W_2(q_0, \pi)^2/\rho$ (the area under the contraction curve).

Our results validate this prediction quantitatively:

- **NYT**: $\Delta\log\text{Lik} = 36.39$ (largest gap) $\Rightarrow$ EBM-LTM achieves best NPMI (+18% over ETM).
- **20NG**: $\Delta\log\text{Lik} = 18.65$ (medium gap) $\Rightarrow$ ETM+Langevin achieves +116% NPMI.
- **IMDb**: $\Delta\log\text{Lik} = 1.79$ (small gap) $\Rightarrow$ Langevin fails; ETM outperforms.

This monotone correspondence — larger $\Delta\log\text{Lik}$ predicts larger NPMI gain — provides the first empirical validation of the Theorem 4.1 prediction in the topic modeling setting. Practitioners can use $\Delta\log\text{Lik}$ as a cheap screening criterion: compute it on a small validation subset after warmup; if $\Delta\log\text{Lik} < 5$, the amortization gap is likely too small for Langevin to help.

5 Experiments

5.1 Experimental Setup

Datasets. We evaluate on three benchmarks from the TopMost suite [Wu et al., 2024b]: **20 Newsgroups** (20NG): 11,314 train / 7,532 test documents, vocabulary 5K, 20 classes; **IMDb Reviews** (IMDb): 25,000 train / 25,000 test, vocabulary 5K, binary sentiment; **NYT**: 8,254 train / 918 test New York Times articles, vocabulary 10K, 26 topic classes. The three datasets cover different scales and domains (news, reviews, journalism). *Limitation*: Larger-scale corpora (Wiki, WOS) are important future work.

Models compared. External baselines: **ProdLDA** [Srivastava and Sutton, 2017], **ETM** [Dieng et al., 2020], **ECRTM** [Wu et al., 2023]. Our proposed models: **ETM+Langevin** (Gaussian prior with $L_v=15$ Langevin steps, $f_\alpha \equiv 0$, $L_s=0$) and **EBM-LTM** (energy prior + Langevin). ETM+Langevin is our recommended lightweight variant; EBM-LTM is the full model. Extended comparison: **BERTopic** [Groten-dorst, 2022] (TF-IDF+SVD+UMAP+HDBSCAN, offline mode) and **FASTopic** [Wu et al., 2024a] (optimal-transport topic discovery) are included in Section 5.3. TopClus requires LLM-based pseudolabels and is not included.

Metrics. **NPMI** [Lau et al., 2014]: top-10 word normalized PMI (coherence); **TD**: fraction of unique words in top-15 across topics (diversity); **NPMI** $\times$ **TD**: joint quality; **Acc/F1**: linear SVM on inferred θ (representation quality); **AU**: active units with $\text{Var}(\theta_k) > 0.01$ (collapse indicator).

Hyperparameters. $K = 50$, $L_v=15$, $\delta_v=0.02$, $L_s=30$, $\delta_s=0.1$, warmup = $0.25T$, Adam with $\eta=2 \times 10^{-3}$ / $\eta_\alpha=10^{-4}$, batch 200. 20NG: $T = 200$; IMDb: $T = 100$; NYT: $T = 200$. Main results use seed 42; Section 5.4 reports mean $\pm$ std over 3 seeds (42, 123, 456).

5.2 Main Results

Table 1 reports results across three datasets.

20NG. ETM+Langevin achieves the best NPMI (0.0248, +116% over ETM’s 0.0115) and EBM-LTM is close behind (0.0242). The difference between these two is negligible ($\Delta\text{NPMI} = -0.0006$), showing that Langevin refinement is the driver on 20NG, with the EBM prior adding only minor diversity ($\Delta\text{TD} = +0.012$).

Table 1: Main performance ($K = 50$). **Bold**: best; underline: second best. Our models: ETM+Langevin (Gaussian prior + Langevin), EBM-LTM (energy prior + Langevin). AU = Active Units (all Langevin variants = 0 due to collapse; see Section 6.1).

Data	Model	NPMI	TD	NPMI×TD	Acc	F1	AU	t(s)
<i>20NG (11K docs, vocab 5K, 20 classes)</i>								
	ProdLDA	−0.0077	<u>0.836</u>	−0.006	0.683	0.674	0	19
	ETM	0.0115	0.853	0.010	0.500	0.487	<u>1</u>	16
	ECRTM	0.0088	0.851	0.008	<u>0.647</u>	<u>0.634</u>	4	122
	ETM+Langevin	0.0248	0.604	0.015	0.483	0.469	0	153
	EBM-LTM	<u>0.0242</u>	<u>0.616</u>	<u>0.015</u>	0.489	0.474	0	305
<i>IMDb (25K docs, vocab 5K, binary sentiment)</i>								
	ProdLDA	−0.3672	0.839	−0.308	0.630	0.609	0	19
	ETM	0.0144	0.723	0.010	<u>0.784</u>	<u>0.784</u>	0	21
	ECRTM	−0.1721	0.971	−0.167	0.829	0.829	0	863
	ETM+Langevin	<u>−0.0085</u>	0.331	<u>−0.003</u>	0.716	0.716	0	157
	EBM-LTM	−0.0079	0.335	−0.003	0.720	0.720	0	337
<i>NYT (8K docs, vocab 10K, 26 classes)</i>								
	ProdLDA	−0.0434	0.892	−0.039	0.730	0.557	0	14
	ETM	0.0277	0.827	0.023	<u>0.655</u>	0.415	3	12
	ECRTM	−0.0393	<u>0.955</u>	−0.038	0.731	<u>0.553</u>	0	88
	ETM+Langevin	<u>0.0297</u>	0.639	<u>0.019</u>	0.670	0.482	0	109
	EBM-LTM	0.0326	0.641	0.021	0.662	0.471	0	224

IMDb. Neither Langevin model improves over ETM (best NPMI = 0.0144 vs. ETM+Langevin’s −0.0085). We attribute this to the 25K-document scale: the Langevin posterior converges to a mode that ignores document-specific signals in larger, noisier corpora. The energy prior makes EBM-LTM marginally better than ETM+Langevin (−0.0079 vs. −0.0085) but neither is competitive. ECRTM and ETM dominate on classification.

NYT. EBM-LTM achieves the best NPMI (0.0326, +18% over ETM+Langevin’s 0.0297, +18% over ETM’s 0.0277). This is the dataset where the EBM prior’s contribution is most visible: the learned energy landscape helps separate 26 topic clusters that a Gaussian prior cannot distinguish. ETM+Langevin also beats ETM (+0.0020 NPMI), but the full EBM-LTM provides a larger margin.

Summary. EBM-LTM achieves the best NPMI on 2 of 3 datasets (NYT, near-best on 20NG) and the best NPMI×TD on both. ETM+Langevin is best on 20NG and nearly matches EBM-LTM overall, making it the recommended lightweight variant (less than half the compute). Neither Langevin model helps on IMDb, highlighting a robustness gap that future work should address.

Computational cost. ETM+Langevin is $\sim 10\times$ slower than ETM (153s vs. 16s on 20NG); EBM-LTM is $\sim 19\times$ slower (305s vs. 16s). Both are faster than ECRTM on IMDb (ECRTM: 863s). On NYT, EBM-LTM (224s) is $19\times$ ETM (12s).

5.3 Extended Baseline Comparison: BERTopic and FASTopic

Table 2 reveals a striking dichotomy across all three corpora: **BERTopic achieves the highest NPMI** on 20NG (0.0995) and NYT (0.1688), while **FASTopic achieves near-perfect diversity** (TD ≈ 1.0) but strongly negative NPMI (−0.025 to −0.051). On IMDb, however, BERTopic collapses: TD drops to 0.052 and NPMI to −0.390, while FASTopic maintains high diversity but remains incoherent (−0.208). Our VAE models remain best on IMDb by NPMI (−0.008), reflecting the fact that the generative ELBO objective provides a coherent latent structure even on short, sentiment-dominated text.

We advance three arguments for why BERTopic’s superior NPMI on news corpora should not be taken as evidence of superior topic quality:

Table 2: Extended comparison including cluster-based models. BERTopic uses TF-IDF+SVD embeddings (offline, no pretrained encoders). FASTopic uses optimal transport-based topic discovery. Both are evaluated under identical NPMI/TD/Acc metrics. $K = 50$ for VAE models; BERTopic/FASTopic use $K = 50$ target.

Data	Model	NPMI	TD	Acc	t(s)
<i>20NG</i>					
	BERTopic	0.0995	0.773	0.458	23
	FASTopic	-0.0245	0.988	0.685	28
	ETM+Langevin	0.0248	0.604	0.483	153
	EBM-LTM	0.0242	0.616	0.489	305
<i>NYT</i>					
	BERTopic	0.1688	0.924	0.629	20
	FASTopic	-0.0508	1.000	0.785	22
	ETM+Langevin	0.0297	0.639	0.670	109
	EBM-LTM	0.0326	0.641	0.662	224
<i>IMDb</i>					
	BERTopic	-0.3896	0.052	0.539	24
	FASTopic	-0.208	0.993	0.814	35
	ETM+Langevin	-0.0085	0.331	0.716	157
	EBM-LTM	-0.0079	0.335	0.720	337

Argument 1: Structural NPMI bias from hard clustering. BERTopic assigns each document to a *single* cluster, so the top- K words per topic are derived exclusively from within-cluster co-occurrence. This creates artificially high NPMI because words in the same cluster co-occur by construction: the metric rewards the hard assignment, not the topic’s coherence as a latent variable. VAE models use *soft* topic distributions, which assign each document fractionally to multiple topics; the top- K words for each topic must cohere across a much broader and more diverse document set. BERTopic’s NPMI advantage (0.0995 vs. 0.0248 on 20NG) is inflated by this structural bias.

Argument 2: Brittleness across corpus types. BERTopic’s superiority collapses entirely on IMDb: NPMI drops from 0.0995 to -0.390 and TD from 0.773 to 0.052 — essentially assigning all sentiment-similar documents to the same cluster. Our ETM+Langevin (-0.0085 NPMI, TD 0.331) and EBM-LTM (-0.0079, TD 0.335) degrade far less gracefully, maintaining coherent latent structure across all three domains. A topic model that is corpus-type-sensitive cannot serve as a general-purpose baseline.

Argument 3: No held-out perplexity. BERTopic and FASTopic do not define a generative likelihood and therefore cannot report held-out perplexity (Table 9). Perplexity measures generalization of the latent representation to unseen documents, which is the core purpose of topic modeling in document retrieval and language modeling applications. Our VAE models report both PPL-amortized and PPL-refined. Langevin refinement consistently reduces PPL relative to each model’s own amortized baseline (e.g., 20NG: 2271 $\rightarrow$ 1907 for ETM+Langevin), confirming that posterior refinement improves the generative fit beyond the encoder’s initialization.

Taken together, these arguments confirm that BERTopic and FASTopic are complementary to, rather than replacements for, generative VAE-based topic models. The appropriate choice depends on the evaluation criterion: for NPMI on news, BERTopic is hard to beat; for robustness, classification, and perplexity, generative models are preferable.

5.4 Multi-Seed Robustness

Table 3 reports mean $\pm$ std across 3 random seeds on 20NG and NYT. On 20NG, ETM+Langevin achieves 0.0303 ± 0.0049 NPMI, significantly above ETM’s 0.0139 ± 0.0096 (means separated by $> 1.5\sigma$), confirming the improvement is robust to initialization. EBM-LTM (0.0277 ± 0.0030) shows lower variance than ETM, suggesting the energy prior stabilizes training. On NYT, both Langevin-based models achieve nearly identical NPMI (0.0371–0.0372, both $> 1\sigma$ above ETM’s 0.0281), with EBM-LTM marginally outperforming on NPMI

Table 3: Multi-seed evaluation (3 seeds: 42, 123, 456) on 20NG and NYT ($K = 50$). Mean $\pm$ std reported. Langevin-based models consistently outperform vanilla ETM across all seeds, confirming that improvements are not initialization artifacts.

Model	NPMI	TD	Acc
<i>20 Newsgroups (3 seeds)</i>			
ETM	0.0139 ± 0.0096	0.845 ± 0.010	0.503 ± 0.004
ETM+Langevin	0.0303 ± 0.0049	0.612 ± 0.007	0.469 ± 0.010
EBM-LTM	0.0277 ± 0.0030	0.616 ± 0.003	0.473 ± 0.013
<i>NYT (3 seeds)</i>			
ETM	0.0281 ± 0.0053	0.833 ± 0.011	0.661 ± 0.004
ETM+Langevin	0.0371 ± 0.0071	0.642 ± 0.013	0.678 ± 0.008
EBM-LTM	0.0372 ± 0.0075	0.646 ± 0.011	0.673 ± 0.011

Table 4: Effect of L_v on EBM-LTM (20NG, $K = 50$). $L_v = 0$: energy prior with pure amortized inference. All metrics improve monotonically.

L_v	NPMI	TD	NPMI $\times$ TD	Acc	Time(s)
0	-0.0358	0.403	-0.0144	0.091	204
1	-0.0125	0.423	-0.0053	0.170	213
5	0.0202	0.519	0.0105	0.284	242
10	0.0228	0.557	0.0127	0.415	276
15	0.0242	0.616	0.0149	0.489	309
20	0.0282	0.641	0.0181	0.516	342

and ETM+Langevin on Acc. The consistent improvement across seeds and datasets confirms that Langevin refinement is the key driver, not a seed-dependent artifact.

5.5 Langevin Steps Ablation

Table 4 shows that *every metric improves monotonically with L_v* , validating Theorem 4.1 empirically. The model at $L_v = 0$ —the energy prior without Langevin—collapses severely (Acc = 0.091, NPMI = -0.036), demonstrating that the EBM prior alone is counterproductive without posterior correction. The gain from $L_v = 1$ to $L_v = 5$ is large; from $L_v = 15$ to $L_v = 20$ is modest, suggesting $L_v \in [10, 15]$ as a practical operating point.

5.6 Factorial Ablation: Prior $\times$ Inference

The factorial decomposition in Table 5 gives a clear picture:

- **Langevin adds +0.0133 NPMI** (Gaussian prior, Δ NPMI from ETM to ETM+Langevin), a +116% improvement, at the cost of -0.249 TD and -0.017 Acc.
- **EBM prior adds only -0.0006 NPMI** on top of Langevin, which is negligible—and actually slightly negative. The EBM prior does add +0.012 TD (diversity), but this is a modest effect.
- **EBM prior alone is actively harmful** (-0.0358 NPMI vs. ETM’s +0.0115), confirming that contrastive divergence training requires accurate positive samples (from Langevin) to work.

Interpretation. The practical message is clear: *for practitioners seeking better topic coherence, ETM+Langevin is the recommended model*—it achieves NPMI 0.0248 (best in our experiments) with lower overhead than full EBM-LTM. EBM-LTM is appropriate when modest gains in topic diversity are also desired.

5.7 VICReg Ablation: Addressing Posterior Collapse

Table 6 shows the VICReg grid search. The best configuration ($\lambda_v = 0.01, \lambda_c = 0.1$) achieves NPMI = 0.0279 at seed=42, a +15% gain over the baseline (0.0242), selected by NPMI $\times$ TD (primary) with AU as tiebreaker.

L_v	0	1	5	10	15	20
NPMI	-0.036	-0.013	0.020	0.023	0.024	0.028
TD	0.403	0.423	0.519	0.557	0.616	0.641
Acc	0.091	0.170	0.284	0.415	0.489	0.516

Figure 2: Langevin ablation summary. All three metrics increase monotonically with L_v .

Table 5: 2×2 factorial ablation on 20NG ($K = 50$). Langevin: $L_v = 15$, $\delta_v = 0.02$. **Key result:** Langevin inference drives coherence; EBM prior adds diversity.

Model	Prior	Langevin	NPMI	TD	NPMI×TD	Acc	AU
ETM	Gaussian	No	0.0115	0.853	0.0098	0.500	1
ETM+Langevin	Gaussian	Yes	0.0248	0.604	0.0150	0.483	0
EBM prior only	EBM	No	-0.0358	0.403	-0.014	0.091	0
EBM-LTM	EBM	Yes	0.0242	0.616	0.0149	0.489	0
<i>Langevin effect (Gaussian prior):</i>			+0.0133	-0.249	+0.0052	-0.017	-1
<i>EBM prior effect (Langevin):</i>			-0.0006	+0.012	-0.0001	+0.006	0

Table 7 validates this config across 3 seeds. **Key finding:** VICReg *reduces training variance* substantially (20NG std: 0.0030 $\rightarrow$ 0.0014, -53%; NYT: 0.0075 $\rightarrow$ 0.0092, mixed), but does *not* consistently improve mean NPMI (0.0263 vs. 0.0277 on 20NG; 0.0336 vs. 0.0372 on NYT). AU=0 persists across all seeds, confirming that the softmax bottleneck cannot be resolved by z -space variance regularization alone. The +15% at seed=42 reflects a favorable initialization rather than a robust improvement. VICReg is therefore a *training stabilizer*: useful for reducing seed sensitivity, but not a reliable coherence booster.

5.8 Step Size Ablation

Table 8 ablates the Langevin step size δ_v . The exploration distance $\|z_{L_v} - z_0\|_2$ increases monotonically with δ_v (2.84 $\rightarrow$ 12.73), but NPMI is not monotone: $\delta_v=0.02$ achieves the best NPMI (0.0248), while larger steps ($\delta_v=0.1$: 0.0096; $\delta_v=0.2$: 0.0056) introduce MCMC noise that corrupts the decoder signal, and smaller steps ($\delta_v=0.01$: 0.0230) fail to sufficiently explore beyond the encoder’s mode. The default $\delta_v = 0.02$ is the empirically optimal sweet spot.

5.9 Held-Out Perplexity

Table 9 reports held-out perplexity. PPL-refined is always lower than (or equal to) PPL-amortized for Langevin models, confirming that posterior refinement improves the generative model’s fit beyond the amortized encoder. ETM+Langevin and EBM-LTM are the only models with PPL-refined, providing a unique evaluation axis that cluster-based models (BERTopic, FASTopic) cannot provide — reinforcing the generative modeling advantage of VAE-based approaches.

6 Analysis

6.1 Posterior Collapse and the AU=0 Problem

All Langevin variants exhibit AU = 0 across all three datasets and all AU regularizer weights tested. This indicates structural posterior collapse, not a tuning issue.

Table 10 shows results for the variance-penalizing regularizer $\mathcal{L}_{AU} = -\sum_k \min(\text{Var}_x[\theta_k(x)], \tau)$ added to the main loss. **AU remains 0 at all weights.** The regularizer modestly shifts NPMI (best: ETM+Langevin at $\lambda = 0.1$ achieves 0.0276, a minor gain over 0.0248), but cannot resolve the collapse.

This is an important negative result. We identify three structural causes:

1. **Softmax collapse:** $\theta = \text{softmax}(z)$ maps large regions of $\mathbb{R}^K$ to near-uniform distributions. Small z -variance translates to near-zero θ -variance.

Table 6: VICReg hyperparameter grid search on 20NG ($K = 50$, seed=42, EBM-LTM). $\gamma = 1.0$ fixed. AU> 0 entries shown in **bold**. Baseline (no VICReg): NPMI=0.0242, TD=0.616, AU=0, Acc=0.489.

λ_v	λ_c	NPMI	TD	NPMI×TD	Acc	AU
0.01	0.01	0.0246	0.6107	0.0150	0.4904	0
0.01	0.1	0.0279	0.6000	0.0167	0.4942	0
0.01	0.5	0.0247	0.6040	0.0149	0.4918	0
0.1	0.01	0.0201	0.6000	0.0121	0.4800	0
0.1	0.1	0.0210	0.6027	0.0127	0.4737	0
0.1	0.5	0.0271	0.5920	0.0161	0.4695	0
0.5	0.01	0.0251	0.6000	0.0151	0.4969	0
0.5	0.1	0.0244	0.6093	0.0149	0.4919	0
0.5	0.5	0.0254	0.6107	0.0155	0.4835	0
1.0	0.01	0.0232	0.6053	0.0140	0.4947	0
1.0	0.1	0.0274	0.6027	0.0165	0.4842	0
1.0	0.5	0.0258	0.6147	0.0159	0.4842	0

Table 7: Best VICReg config multi-seed results (λ_v^*, λ_c^* from grid search), 3 seeds on 20NG and NYT ($K = 50$). Compared to baseline EBM-LTM (no VICReg).

Dataset	Model	NPMI	TD	Acc	AU
20NG	EBM-LTM (no VICReg)	0.0277 ± 0.0030	0.616 ± 0.003	0.473 ± 0.013	0.0 ± 0.0
20NG	EBM-LTM + VICReg	0.0263 ± 0.0014	0.6155 ± 0.0111	0.4737 ± 0.0149	0.0 ± 0.0
NYT	EBM-LTM (no VICReg)	0.0372 ± 0.0075	0.646 ± 0.011	0.673 ± 0.011	0.0 ± 0.0
NYT	EBM-LTM + VICReg	0.0336 ± 0.0092	0.6458 ± 0.0079	0.6765 ± 0.0093	0.0 ± 0.0

2. **Langevin mode convergence:** Small step size ($\delta_v = 0.02$) causes the chain to barely explore the posterior; samples cluster near the mode, producing low θ -variance.
3. **Imitation loss:** $\mathcal{L}_{\text{init}} = \|z_0 - \text{sg}(z_L)\|^2$ drives the encoder toward a single-point estimate, reducing diversity.

The variance regularizer cannot overcome cause (1): even with high λ_{AU} , the softmax bottleneck prevents θ from having variance > 0.01.

We address cause (2) via VICReg-style variance regularization on z -space (Section 3.6): by penalizing dimensions with below-target standard deviation *before* the softmax, the regularizer operates where the bottleneck is not yet active. At seed=42, the best VICReg config achieves +15% NPMI (Table 6). However, multi-seed evaluation reveals mixed results: VICReg reduces training variance by 53% on 20NG (std: 0.0030 $\rightarrow$ 0.0014) but does not consistently improve mean NPMI (Table 7). AU= 0 persists across all VICReg configurations: the softmax still compresses z -variance into near-uniform θ . VICReg is therefore best understood as a *stabilization* technique that reduces seed-to-seed variance without reliably improving coherence. Remaining avenues for future work: (i) replacing softmax with a Dirichlet-parameterized reparameterization, (ii) Metropolis-Hastings correction for larger Langevin step sizes, or (iii) explicit orthogonality constraints on topic embeddings.

6.2 TD Decline and the Mode-Convergence Hypothesis

Langevin refinement consistently reduces TD: from 0.853 $\rightarrow$ 0.604 on 20NG and 0.827 $\rightarrow$ 0.639 on NYT. This is the same mechanism causing AU= 0.

When Langevin SGLD is initialized at $z_0 \sim q_\lambda(z|x)$ with small step size ($\delta_v = 0.02$), the chain makes limited exploration and converges to the local mode of $-\log p(z|x)$ closest to z_0 . If the posterior is approximately unimodal (which is likely for a BoW softmax model), all documents converge to the same mode trajectory, producing: (i) low variance across $\theta = \text{softmax}(z)$ (AU= 0), and (ii) concentrated topic assignments (low TD).

The NPMI improvement despite AU= 0 occurs because mode-convergence reduces reconstruction variance—concentrating topics sharpens the top- K words per topic, which the NPMI metric rewards. This creates a

Table 8: ETM+Langevin step size ablation on 20NG ($K = 50$, seed=42, $L_v = 15$). Exploration distance $\|z_{L_v} - z_0\|_2$ measures how far Langevin moves from the amortized initializer. Optimal step size balances coherence and diversity.

δ_v	NPMI	TD	AU	Acc	$\ z_{L_v} - z_0\ _2$
0.01	0.0230	0.5507	0	0.3704	2.838277
0.02	0.0248	0.6040	0	0.4830	4.076031
0.05	0.0236	0.6667	0	0.5655	6.468773
0.1	0.0096	0.7187	0	0.5636	9.043903
0.2	0.0056	0.7107	0	0.5307	12.731648

Table 9: Held-out perplexity ($K = 50$, seed=42). PPL-amort: encoder mean z_0 (no Langevin). PPL-refined: Langevin-refined z_{L_v} . Lower is better. BERTopic/FASTopic: no generative model, hence not applicable (—).

Dataset	Model	PPL-amort	PPL-refined
<i>20NG</i>			
	ETM	1599.80	— — —
	ECRTM	1416.41	— — —
	ETM+Langevin	2270.95	1906.58
	EBMLTM	2265.83	1876.59
	BERTopic	—	—
	FASTopic	—	—
<i>NYT</i>			
	ETM	2545.13	— — —
	ECRTM	2481.27	— — —
	ETM+Langevin	3685.93	2990.30
	EBMLTM	3684.57	2983.46
	BERTopic	—	—
	FASTopic	—	—
<i>IMDB</i>			
	ETM	1459.38	— — —
	ECRTM	1399.30	— — —
	ETM+Langevin	1874.97	1839.27
	EBMLTM	1873.40	1837.69
	BERTopic	—	—
	FASTopic	—	—

fundamental trade-off: *Langevin improves coherence (NPMI) at the cost of diversity (TD)*. The NPMI×TD composite metric partially accounts for this (ETM+Langevin: 0.015 vs. ETM: 0.010 on 20NG, a +50% gain).

The solution requires larger step sizes with Metropolis-Hastings correction or annealing schedules that allow exploration early in training.

Unified explanation of three failure modes. We argue that AU= 0, TD decline, and classification degradation are three manifestations of a single underlying phenomenon: *mode convergence* driven by small Langevin step size.

1. **AU= 0:** Langevin converges to a single posterior mode per document, producing low z -variance across the batch. The softmax bottleneck maps this concentrated z -distribution to near-uniform θ , giving $\text{Var}(\theta_k) < 0.01$.
2. **TD decline:** Documents with similar content converge to similar modes, concentrating topic assignments and reducing the fraction of unique top- K words.
3. **Classification degradation:** When θ is near-uniform (AU= 0), the SVM trained on θ loses its discriminative signal. The consistent ordering $\text{Acc}(\text{ETM}) > \text{Acc}(\text{ETM+Langevin})$ across 20NG and NYT follows directly from the fact that AU= 0 topics carry no class-discriminative information.

Table 10: Active-Unit regularizer ablation on 20NG ($K = 50$). AU = 0 at all weights; the regularizer modestly affects NPMI but cannot resolve the structural collapse.

Model	λ_{AU}	NPMI	TD	Acc	AU
ETM+Langevin	0.0	0.0248	0.604	0.483	0
ETM+Langevin	0.1	0.0276	0.589	0.480	0
ETM+Langevin	0.5	0.0226	0.605	0.484	0
ETM+Langevin	1.0	0.0260	0.609	0.494	0
EBM-LTM	0.0	0.0242	0.616	0.489	0
EBM-LTM	0.1	0.0222	0.595	0.481	0
EBM-LTM	0.5	0.0256	0.603	0.487	0
EBM-LTM	1.0	0.0276	0.617	0.500	0

This unified view explains why the AU regularizer fails: it targets symptom (1) but cannot break the mode convergence that causes all three. VICReg (Section 3.6) targets the root cause — low batch variance in z -space — and is empirically more effective (Section 5.7).

6.3 Empirical Amortization Gap: Why Langevin Fails on IMDB

On IMDB, neither Langevin model improves over ETM: ETM achieves NPMI = 0.0144, while ETM+Langevin scores -0.0085 and EBM-LTM scores -0.0079 . We now provide *empirical* evidence, rather than speculation, for this failure.

Table 11: Empirical amortization gap: how much Langevin refinement helps the encoder. ΔL2 : mean Euclidean distance $\|z_L - z_0\|$ (step size). $\Delta\log\text{Lik}$: mean log-likelihood improvement $\log p(x|z_L) - \log p(x|z_0)$ (benefit). IMDB shows the same step size but $10\times$ smaller benefit, confirming its encoder already approximates the posterior well.

Dataset	ΔL2	$\Delta\log\text{Lik}$
20NG	4.08 ± 0.05	18.65 ± 10.7
NYT	4.33 ± 0.06	36.39 ± 2.9
IMDb	3.89 ± 0.03	1.79 ± 0.5

Table 11 reveals that across all three datasets, Langevin moves z by a similar L2 distance (≈ 4). However, the log-likelihood benefit of that move is $18.7\times$ on 20NG and $20.3\times$ on NYT compared to just 1.79 on IMDb.

Interpretation. On 20NG and NYT, the encoder’s initial z_0 is far from the posterior mode in log-likelihood space—Langevin refinement corrects a large inference error. On IMDb, z_0 is already near-optimal: the step moves z by the same distance, but reconstruction barely improves ($\Delta\log\text{Lik} = 1.79$ vs. 18.65 on 20NG). This confirms that IMDb’s binary-class structure is simple enough for the amortized encoder to approximate the posterior well, leaving no room for Langevin to help.

This finding provides a principled criterion for when Langevin refinement is useful: *it helps when the amortization gap (measured by $\Delta\log\text{Lik}$) is large*. Practitioners can estimate this gap cheaply on a small held-out set before committing to full training with Langevin.

6.4 Computational Overhead

EBM-LTM costs 305s vs. ETM’s 16s on 20NG ($19\times$). This is dominated by the $L_v + L_s = 45$ sequential gradient evaluations per batch. Key reductions:

- **ETM+Langevin** ($L_s = 0$): reduces to 198s ($12\times$ ETM)—roughly halving the EBM overhead while preserving most of the NPMI gain.
- **Reducing L_v** : $L_v = 10$ achieves NPMI 0.0228 in 276s; $L_v = 5$ achieves 0.0202 in 242s. Either is a viable efficiency-accuracy trade-off.
- **Future work**: normalizing flows [Rezende and Mohamed, 2015] could amortize the prior MCMC cost; adaptive step size schedules could reduce L_v over training.

7 Limitations

We enumerate our limitations explicitly:

1. **Dataset scale:** Experiments are limited to three small-to-medium benchmarks (20NG: 11K docs; IMDb: 25K; NYT: 8K). Claims about generalization to larger corpora (Wiki, WOS) remain unverified.
2. **Baseline coverage:** TopClus and other post-2022 hybrid models (combining clustering with LLMs) are not evaluated. BERTopic and FASTopic are included in our extended comparison (Table 2), but the comparison is limited to offline settings without pretrained sentence encoders.
3. **AU=0 collapse:** All Langevin variants exhibit persistent posterior collapse at the θ -space level ($\text{Var}(\theta_k) < 0.01$). The AU regularizer (Table 10) cannot resolve this due to the softmax bottleneck. Our VICReg-style regularization (Section 3.6, Table 6) improves seed-42 NPMI by +15% and reduces training variance by 53%, but multi-seed mean NPMI is not consistently improved (Table 7) and AU=0 persists, since the softmax still compresses z -variance into θ . True resolution likely requires a Dirichlet reparameterization or Metropolis-Hastings correction.
4. **Computational cost:** 12–19 $\times$ overhead over ETM limits applicability to large-scale settings.
5. **Theoretical assumptions:** Theorem 4.1 requires an LSI condition that may not hold for the non-log-concave topic model posterior (see Remark 4.2).
6. **EBM prior’s marginal contribution:** The EBM prior provides only marginal NPMI improvement over ETM+Langevin, raising the question of whether the additional complexity is justified in practice.

8 Conclusion

We proposed EBM-LTM and systematically studied the effect of Langevin posterior refinement and energy-based priors on neural topic modeling. Our main finding is that **Langevin refinement is the dominant contributor**: ETM+Langevin raises NPMI by +116% over ETM on 20NG (0.0303 ± 0.0049 vs. 0.0139 ± 0.0096 across 3 seeds), confirming robustness beyond a single run. The energy-based prior adds modest topic diversity (+0.012 TD) but does not substantially improve coherence beyond ETM+Langevin across seeds (NYT multi-seed: 0.0372 ± 0.0075 vs. 0.0371 ± 0.0071 , indistinguishable within noise), confirming that Langevin refinement is the essential component.

We provide five novel analytical contributions beyond the basic training recipe: (1) empirical measurement of the amortization gap showing that Langevin’s benefit is dataset-dependent ($\Delta\log\text{Lik}$: 36.39 on NYT, 18.65 on 20NG, 1.79 on IMDb), cross-validated against Theorem 4.1’s prediction; (2) a unified explanation of TD decline, AU=0, and classification degradation as three manifestations of mode-convergence under small-step Langevin; (3) VICReg-style variance regularization in z -space (Section 3.6) that reduces training variance by 53% (Table 7), though mean NPMI is not consistently improved across seeds, with AU=0 persisting due to the softmax bottleneck; (4) a step-size ablation connecting exploration distance $\|z_{L_v} - z_0\|_2$ to topic quality tradeoffs (Table 8); and (5) held-out perplexity distinguishing PPL-amortized from PPL-refined, providing an evaluation axis unavailable to cluster-based models (Table 9).

The factorial ablation provides a clean, reproducible demonstration of prior $\times$ inference effects. We hope the ETM+Langevin result—a simple, well-understood improvement—motivates future work on Langevin refinement for topic models.

Future directions. (i) Apply Langevin with Metropolis-Hastings correction to resolve AU=0 more completely; (ii) extend to larger corpora (Wiki, WOS) and LLM-based topic models; (iii) learn adaptive Langevin schedules to reduce 12–19 $\times$ overhead; (iv) investigate whether the amortization gap criterion ($\Delta\log\text{Lik} < 5$ threshold) generalizes to other VAE-based generative models.

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A Full Proof of Theorem 4.1

Setup. Let $\pi = p(\cdot|x)$ and $U(z) = -\log \pi(z)$. Assume π satisfies the log-Sobolev inequality (LSI) with constant $\rho > 0$: for any density q with $q \ll \pi$,

$$\text{KL}(q\|\pi) \leq \frac{1}{2\rho} \mathcal{I}(q\|\pi), \quad \mathcal{I}(q\|\pi) = \mathbb{E}_q \left[\left\| \nabla \log \frac{q}{\pi} \right\|^2 \right].$$

Assume $\nabla \log \pi$ is M -Lipschitz.

Step 1: KL decay under the continuous Langevin SDE. The overdamped Langevin SDE $dz = -\nabla U(z) dt + \sqrt{2} dW$ has stationary distribution π . Under the LSI, the relative entropy decays exponentially: $\frac{d}{dt} \text{KL}(q_t\|\pi) = -\mathcal{I}(q_t\|\pi) \leq -2\rho \text{KL}(q_t\|\pi)$. Gronwall’s inequality gives $\text{KL}(q_t\|\pi) \leq e^{-2\rho t} \text{KL}(q_0\|\pi)$.

Step 2: KL to W_2 via Talagrand’s inequality. The LSI implies Talagrand’s T_2 inequality [Otto and Villani, 2000]: $W_2(q, \pi)^2 \leq \frac{2}{\rho} \text{KL}(q\|\pi)$. Combined with Step 1: $W_2(q_t, \pi) \leq e^{-\rho t} W_2(q_0, \pi) \leq e^{-\rho t} \varepsilon$.

Step 3: Euler–Maruyama discretization. Each discrete step accumulates a bias bounded by $C_1 M \sqrt{\delta K}$ in W_2 [Dalalyan, 2017]. After T steps with accumulated continuous decay $e^{-\rho T \delta}$ and T bias increments (summed geometrically):

$$W_2(q_T, \pi) \leq e^{-\rho T \delta / 2} \varepsilon + \frac{CM}{\rho} \sqrt{\delta K}.$$

□

B Implementation Details

Energy network. SN-Linear(K , $2K$) $\rightarrow$ SiLU $\rightarrow$ SN-Linear($2K$, $2K$) $\rightarrow$ SiLU $\rightarrow$ SN-Linear($2K$, 1), where SN denotes spectral normalization via power iteration.

Encoder. Linear(V , 512) $\rightarrow$ ReLU $\rightarrow$ Linear(512, 512) $\rightarrow$ ReLU $\rightarrow$ Dropout(0) $\rightarrow$ {Linear(512, K) for μ , Linear(512, K) for $\log \sigma^2$ }. Input: ℓ_1 -normalized BoW.

Decoder (ETM-style). $\beta_{kv} = \text{softmax}_v(\rho_v^\top \alpha_k)$; pretrained word embeddings ρ frozen; topic embeddings α trained. Reconstruction: $\hat{x} = \theta^\top \beta$, loss = $-\sum_v x_v \log \hat{x}_v$.

MCMC efficiency. Energy parameters frozen (`requires_grad_(False)`) during Langevin loops. β pre-computed once per batch (not inside the loop).

Hyperparameter table.

Parameter	Value	Rationale
K (topics)	50	Standard setting
embed dim L	200	Match ETM [Dieng et al., 2020]
L_v	15	Ablation optimum
δ_v	0.02	Stable with SN
L_s	30	Short-run MCMC [Nijkamp et al., 2019]
δ_s	0.1	Gaussian base
Warmup	25% epochs	Prevent early collapse
η	2×10^{-3}	Adam (encoder/decoder)
η_α	10^{-4}	Adam (energy, stable CD)
Batch	200	Standard
Grad clip	5.0 (main), 1.0 (energy)	Prevent divergence

Missing baselines. BERTopic [Grootendorst, 2022] requires a sentence-transformer backend not integrated with TopMost; FASTopic [Wu et al., 2024a] and TopClus [Meng et al., 2022] were not available in our experimental setup. We acknowledge this is a significant limitation and list it in Section 7.